A certify-or-reject check for GENCO's N-1 power-flow predictions

First results on the published Texas 2000-bus contingency set. Work in progress.

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When GENCO predicts the grid after an outage, which predictions can someone use without re-running a full study?

- GENCO's predictions are accurate on average. Over these 989 scenarios the mean voltage error is 1.2e-04 pu, the mean angle error 0.25°, and the median power-balance mismatch 0.013 pu.
- In power-flow mode nothing in the model enforces voltage, thermal or generator reactive limits, so an accurate prediction can still describe an operating point that isn't allowed.
- What a user needs per scenario is a yes or no they can rely on: is this an AC-feasible operating point that satisfies every limit, and if not, why.

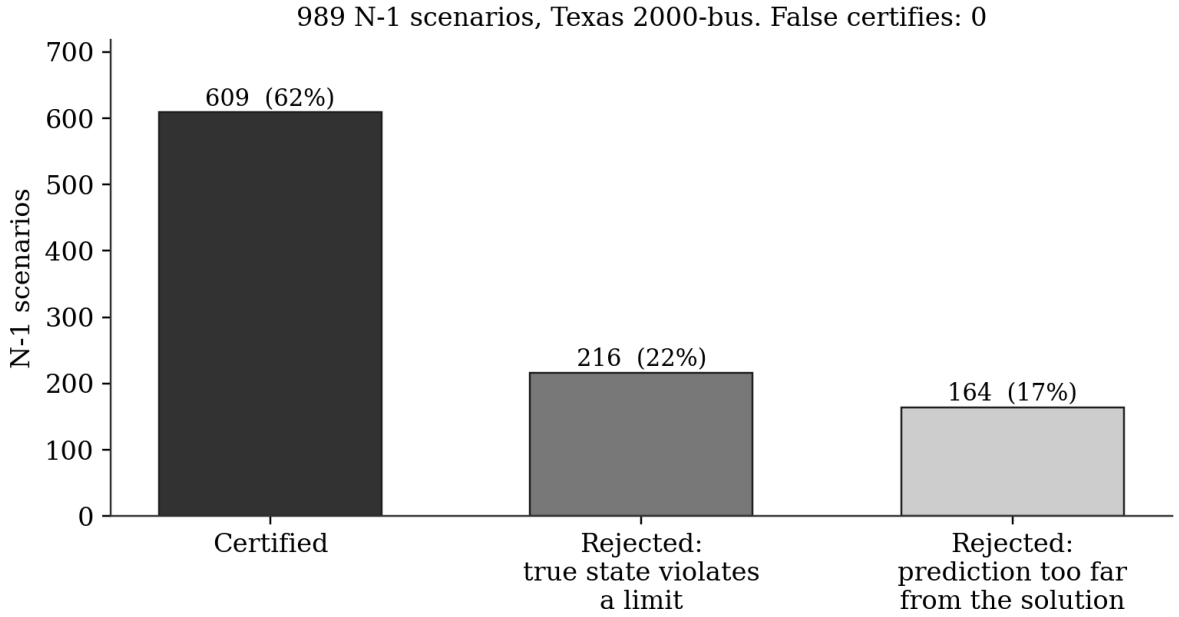
Everything comes from the public GENCO release, so nothing is retrained.

- **Predictions:** the paper checkpoint's N-1 predictions, `huggingface.co/gridfm/genco-pf-contingency`, file `mlflow/eval/test_1/predictions.parquet`.
- **Networks and ground truth:** the published bus, branch, generator and Y-bus tables for the same split, used to rebuild each scenario's network.
- **Scenarios so far:** 5 partitions of `test_1`, 989 N-1 cases on the ACTIVSg2000 (Texas) grid, out of 8,823 in the split.
- **Check before trusting the setup:** the published ground-truth solution satisfies the rebuilt power-flow equations on every scenario, with a worst mismatch around 1e-9 pu.

Solve from the prediction first, then check limits at the solution.

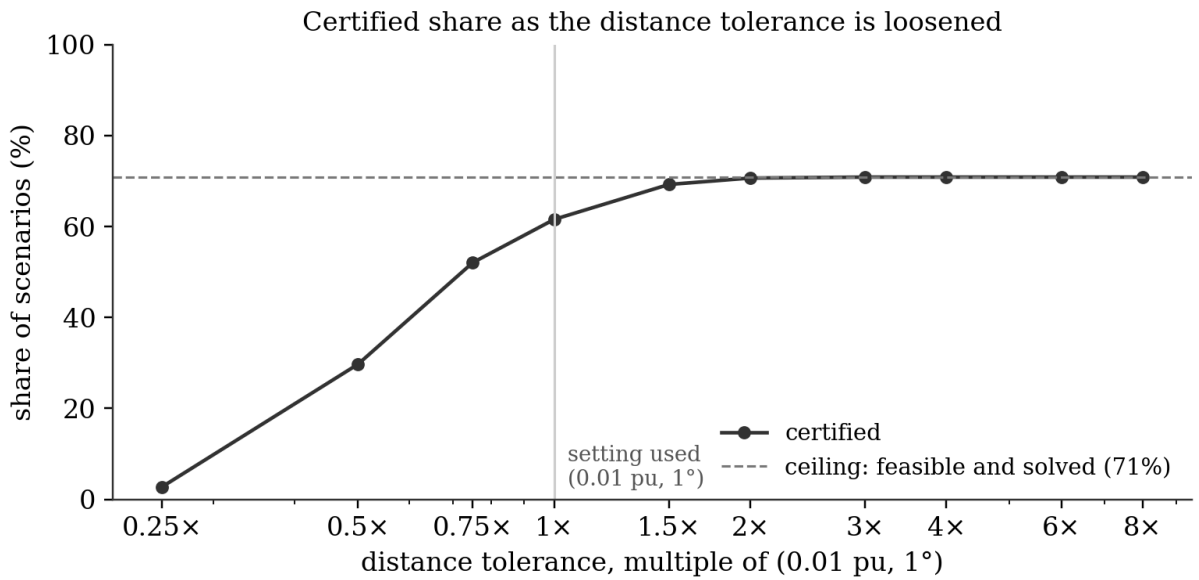
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1  Newton-Raphson, warm-started from GENCO's (Vm, θ), with PV and reference set points fixed.
   Converged means a mismatch below 1e-8 pu within 15 iterations.
2  Distance: the converged point must be within 0.01 pu and 1° of the prediction at every bus.
3  Limits at the converged point: bus voltage bounds, branch MVA ratings at both ends,
   and generator reactive limits aggregated per bus.
4  CERTIFIED only if 1 to 3 pass. Otherwise REJECTED with the reason, and the case goes to a full study.
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- Model-agnostic, plain numpy and scipy, about 300 lines.
- Every certified verdict is audited against the published ground truth. A false certify is a certified case whose true solution differs from the converged point or violates a limit.

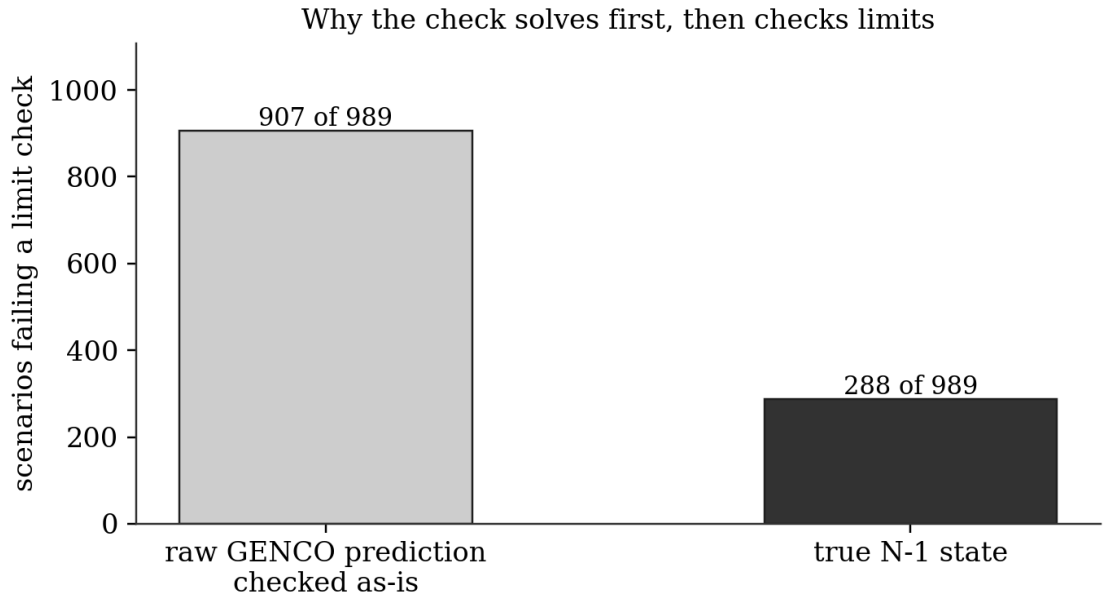


OUTCOME	SCENARIOS
Certified	609 (62%)
Rejected, true state violates a limit	216
Rejected, prediction too far from solution	164
False certifies	**0**

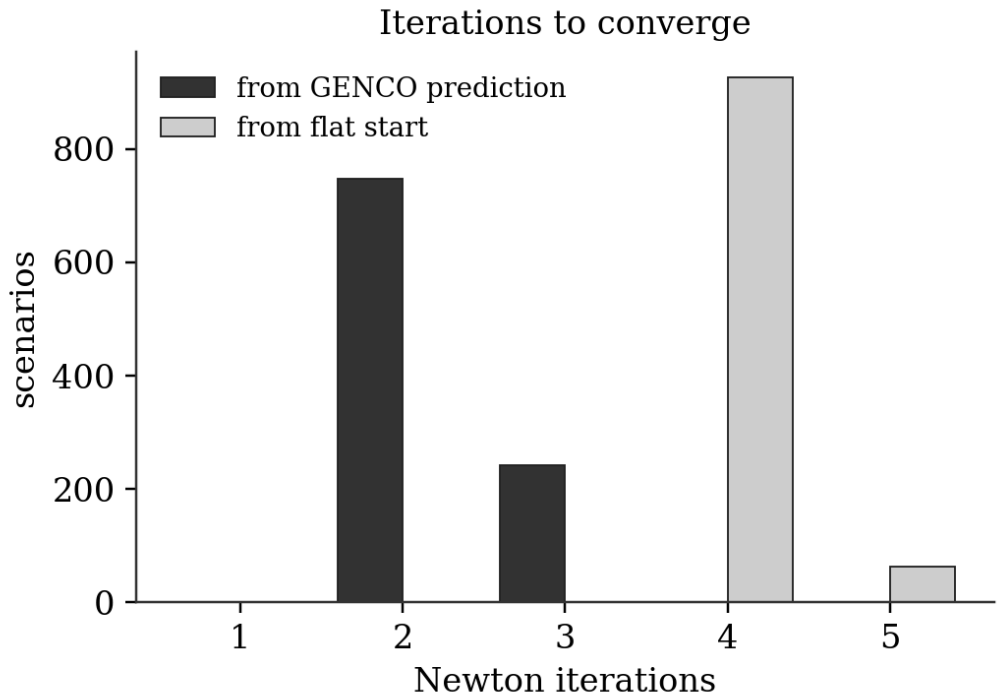
Of the 989 true N-1 states, 288 violate at least one limit, and the check rejected all of them.

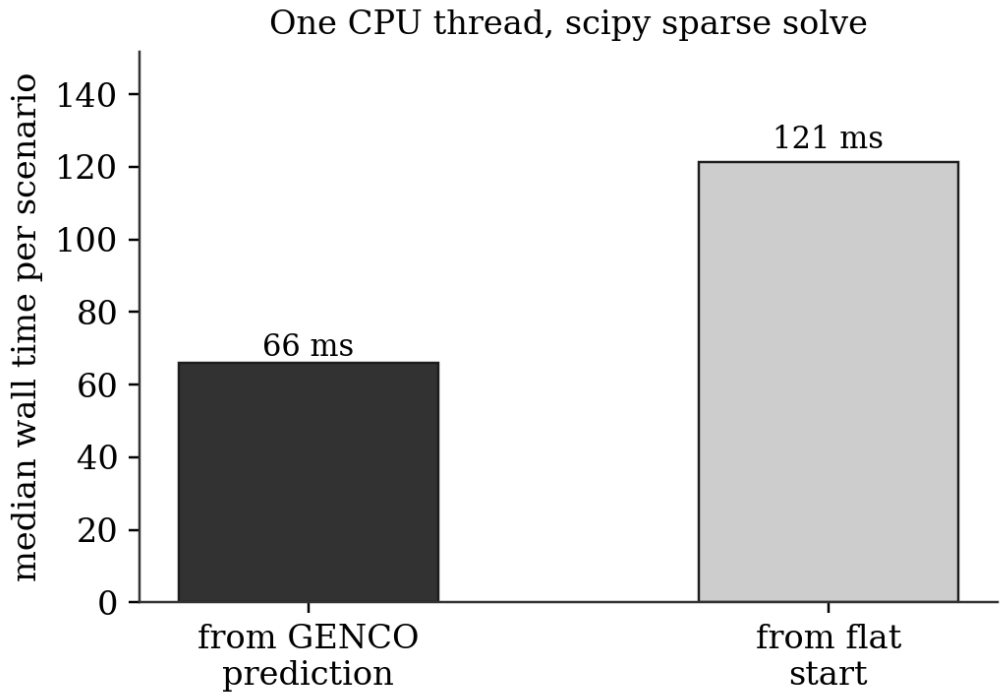


- Most of the "too far" rejections come from the 0.01 pu, 1° rule, which is a design choice.
- Doubling it raises the certified share from 62% to 71%, and it levels off at 71%, the share of scenarios that are feasible.
- Here, whenever Newton converged it converged to the true solution, so loosening the rule didn't add wrong answers. That's a property of this data and I wouldn't assume it carries to harder splits.



- Checked as-is, 907 of 989 raw predictions fail a limit, against 288 of 989 true states.
- Most of the extra failures are generator reactive limits. Reactive power recovered from an approximate state is very sensitive to small voltage errors, so checking the raw prediction rejects cases that are actually fine.
- Solving from the prediction removes that noise, and the prediction still pays off as a warm start.





- Warm-starting from GENCO takes a median of 2 Newton iterations against 4 from a flat start, 66 ms against 121 ms per scenario. No flat-start solve failed on these partitions.

Limits of this first version

- **Coverage:** 989 of 8,823 N-1 scenarios, and none of the N-2 to N-20 splits yet. The pipeline is the same for all of them.
- **Limits checked:** only those in the published tables. Generator reactive limits are aggregated per bus, and angle-difference limits aren't checked.
- **Tolerances:** the distance rule is my choice, and the audit against ground truth is what keeps any setting honest.
- **Not integrated:** it runs on GENCO's saved predictions, not inside gridfm-graphkit.

Next steps, roughly in this order

- Run all partitions of the N-1 split, then the N-2 and higher splits, where I expect predictions and Newton convergence to get harder and the audit to matter more.
- A thin adapter so `gridfm_graphkit predict` outputs go straight into the check and come back with a verdict, a reason and a margin per scenario.
- Per-generator reactive limits and angle-difference limits.
- A feasibility margin per scenario, so a user can see how close a certified case is to a limit, not only whether it passed.

Where I'd value your input

- Which limits and which tolerance would be most useful to OpenGridFM users?
- Is the Texas contingency set the right first benchmark, or is there a case you'd rather see, for example one closer to Hydro-Québec's network?
- Would a post-processing step after `predict` be the right place for this in the codebase?

Code and per-scenario results: github.com/sehajr-singhs/genco-certify